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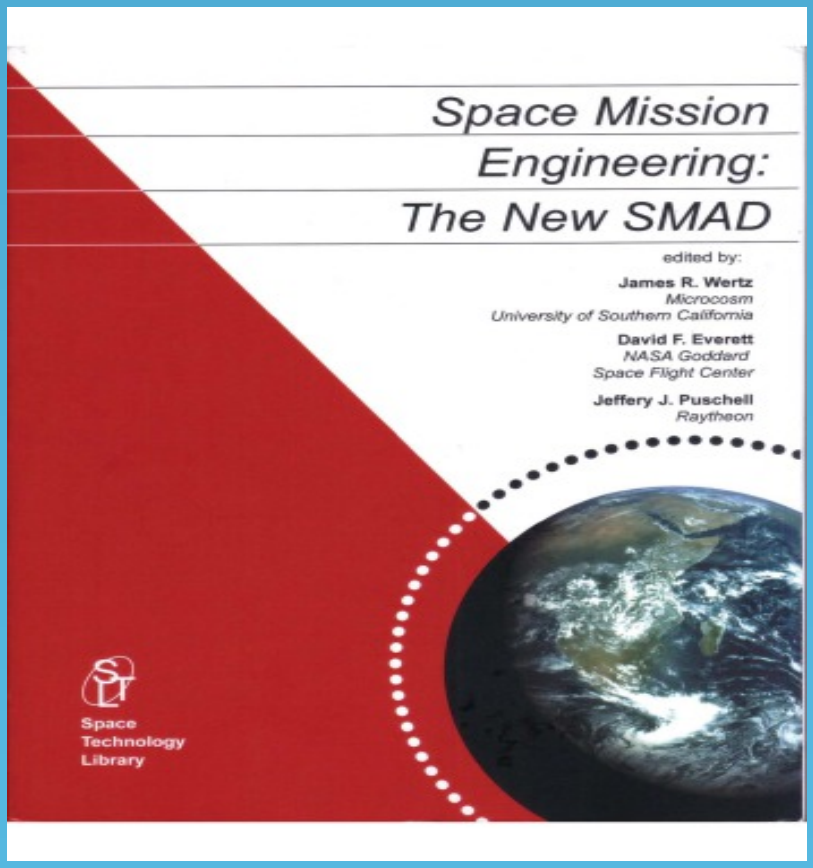

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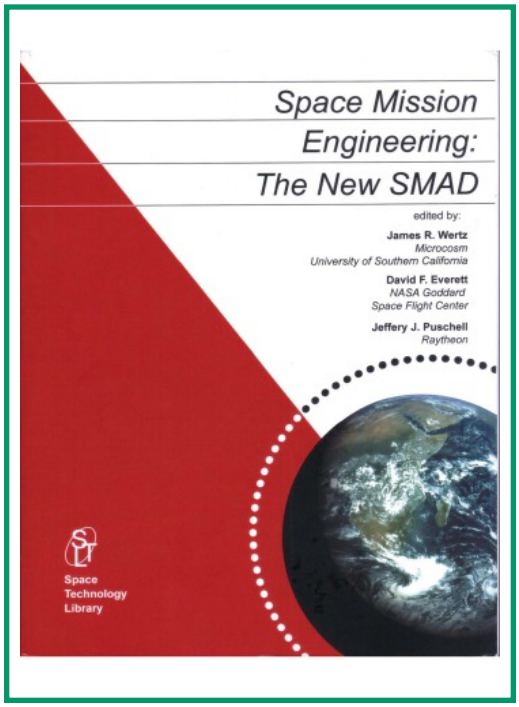

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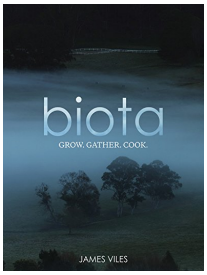

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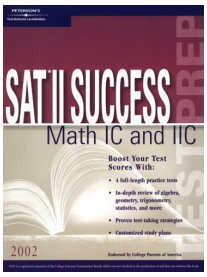

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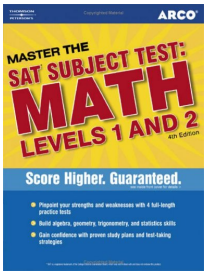

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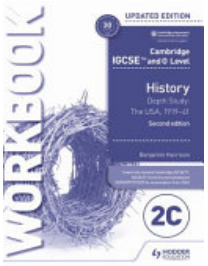

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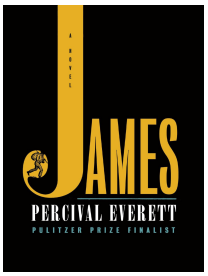

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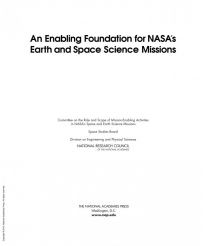

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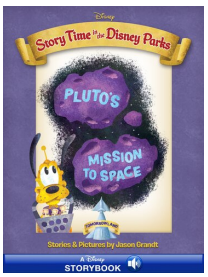

Space Mission Engineering: The New SMAD

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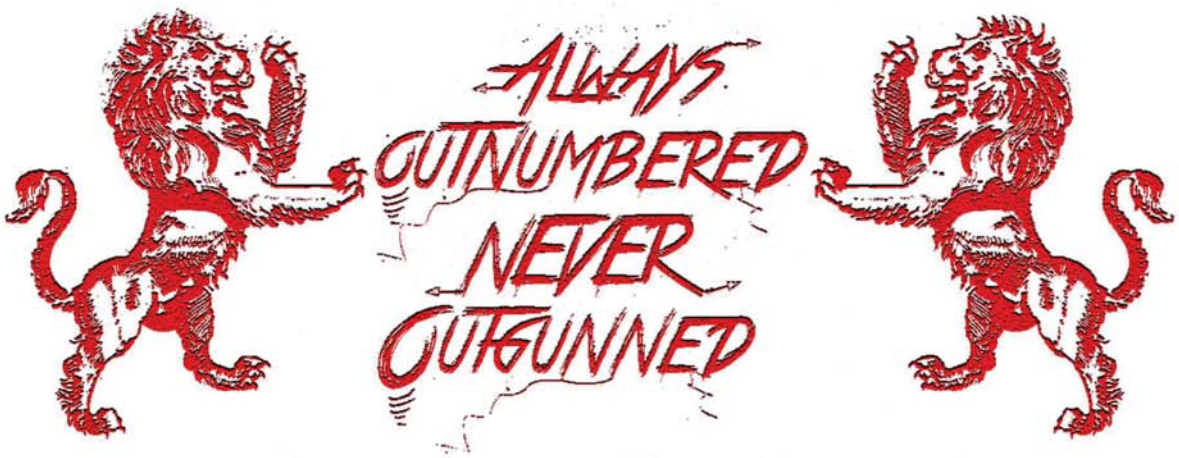

In memory of John

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Units

Symbol	Measurement
A	ampere
AU	astronomical unit
C	coulomb
C	celsius
dB	decibel
deg	degree (angular)
eV	electron volt
Hz	hertz
g	gram
J	joule
K	kelvin
kg	kilogram
km	kilometer
m	meter
N	newton
Ω	ohm
rad	radian
s	second
sec	second
sr	steradian
Pa	pascal
T	tesla
V	volt
W	watt
yr	year

The Greek Alphabet		
Alpha	Α	α
Beta	Β	β
Gamma	Γ	γ
Delta	Δ	δ
Epsilon	Ε	ε
Zeta	Ζ	ζ
Eta	Η	η
Theta	Θ	θ
Opta	Ι	ι
Kappa	Κ	κ
Lambda	Λ	λ
Mu	Μ	μ
Nu	Ν	ν
Xi	Ξ	ξ
Omicron	Ο	ο
Pi	Π	π
Rho	Ρ	ρ
Sigma	Σ	σ
Tau	Τ	τ
Upsilon	Υ	υ
Phi	Φ	φ
Chi	Χ	χ
Psi	Ψ	ψ
Omega	Ω	ω

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Preface

This book is intended as a replacement for *Space Mission Analysis and Design*, known as SMAD to a great many engineers and students. SMAD was first published in 1991 and served as a practical and widely used reference and text for more than 20 years with over 65,000 copies in print. Nonetheless, the worlds of space exploration, space usage, and how books are used and published have all changed in the intervening period, and it is time for a new, unique, and hopefully more useable and useful reference and text in *Space Mission Engineering (SME)*—i.e., defining the parameters and trading on requirements so as to meet the broad, often poorly defined, objectives of a space mission at minimum cost and risk. This is a definition that was introduced in the first edition of SMAD to try and distinguish mission engineering from the more formal process of space systems engineering which had, in many cases, become simply a process of keeping track of all of the requirements and constraints and determining how each was to be met and verified. Certainly both processes are needed, but our emphasis here is on how to go about starting with a blank sheet of paper and **creating** a space mission to meet the needs of the end user as economically and quickly as possible.

It is clear that the technology exists today, or can be created, to do essentially anything we choose to do in space. The major challenges facing us are that space has become dramatically expensive and far too slow to achieve the objectives set for it, and, as a result, is far less useful to the end user than it should be. President Kennedy's call for landing a man on the Moon was followed 8 1/2 years later by the Apollo 11 mission, including developing from scratch the Saturn V launch vehicle—the largest, most reliable, and lowest cost per pound launcher ever built. Today the Saturn V no longer exists and we most likely couldn't finish doing the system design and verification needed for a Moon mission in 8.5 years, even if we could afford to go there.

Fortunately, the space business is beginning to change again, at least in some areas. SmallSats and CubeSats, with dramatically lower cost and shorter schedules, are starting to be taken seriously. Dr. Pedro Rustan [2004], at the time the Head of Advanced Science and Technology for the National Reconnaissance Office, pointed out in 2004 that the goal for space mission engineering should not be just creating low-cost, responsive small satellites, but using the same or comparable techniques to dramatically reduce the cost and schedule for all space systems, including those that are inherently very large and complex.

As a consequence of the need to change the way we do business in space, SME has two complimentary objectives: (a) to explain how and why space missions are built the way they are and (b) to explain how it can be done "faster, better, cheaper." The difficulty of doing this is best captured by the experience of Dan Goldin, the Director of NASA from 1992 to 2001, who introduced this mantra to try to convince the space community that change was critical. After all, almost every other aspect of modern technology has become faster, better, and cheaper in its implementation—including computers, communications, television, automobiles, and aircraft. But culture is very hard to change. Today, if you say "faster, better, cheaper" in a meeting of aerospace engineers, you will almost certainly hear several people respond with "faster, better, cheaper—pick any two." This suggests, of course, that in the view of many space engineers faster, better, and cheaper isn't possible—that space technology is, in effect, as good as it is ever going to get and improvements in one area will mean doing more poorly in some other area. This is the real challenge for SME—to show both where we are today (and how to build space missions according to today's rules) and show clearly that the goals of Dan Goldin, Pete Rustan, and many others can indeed be met as technology advances. It is also our goal to explain how this can be accomplished in as specific and practical terms as possible. Our objective for SME is to satisfy both needs. We would much appreciate feedback from readers about whether we have been successful or how we could do it better. (Send E-mails to bookproject@smad.com or to any of the editors listed below.)

One of the major changes for this volume is that it is both a physical book and a website, www.sme-smad.com. (The password is ASTRODATA and isn't case sensitive.) The website is not intended as a stand-alone site, but as an extension of this volume. It contains articles which go into more detail on many topics, downloadable Excel models for most of the mathematical tables and plots, data tables that can be updated from time-to-time, and a complete listing of all of the references with an annotated bibliography. Nearly all of the references and bibliography have a direct link to where the material can be downloaded for free or, for copyright material, where it can be purchased and downloaded, or where books can be purchased and shipped with next day delivery. Nearly all the sample plots and tables can be customized to your particular space mission. Our goal here is to make the book fit your mission as much as possible and let you find additional details and more in-depth assessments of specific topics within a day or less. We hope it works well and would appreciate your suggestions on how to make it work better.

It is very difficult to express our full appreciation for over 120 authors, reviewers, and advisory committee members, all of whom worked dramatically hard with no contract, no compensation, and no reward other than seeing the work done and a copy of the book when it was complete. These are among the most experienced and busy individuals in the community, giving up free time to provide this material for you, the reader. Without them, it would not have been possible. We are truly

indebted to them for putting in many more hours than most of them had anticipated. Among the reviewers, we would like to especially thank Steve Raque, Mike Swartwout, Jin Kang, Tom Bauer, and Markus Rufer, who reviewed many chapters and made major contributions to the success of the volume. A special thank you goes to Goddard Space Flight Center for supplying many authors and reviewers, and considerable technical data as well.

The work of creating the camera-ready copy fell to the Microcosm Publications Department, under the direction of Chris Maruyama and Donna Klungle. Jenny Greer and Julie Jackson did much of the 2-column formatting. Nearly all of the art work, including the color version of all of the figures and tables for the educator package, was created by Chris Maruyama. Most of the background work, references, bibliography, web references, and Excel models have been created under the direction of Nicola Sarzi-Amade by Nicola, Anthony Shao, Becky Christofferson, and Kyungmo Koo.

As anyone who has undertaken projects of this sort knows, one of the largest contributions comes from the families of the three editors. We greatly appreciate their forbearance and patience with a project that took longer than we hoped and far more evening and weekend hours than any of us would have guessed or wished.

Ultimately, if this book belongs to anyone, it is Pam Lindbloom-Esquinca, the manager of the Microcosm bookstore, and the force behind creating this volume. It was Pam that told us very forcefully to go out and do it, put up several years of bookstore profits to create the camera-ready copy, pushed and cajoled all of the authors, reviewers, and editors, and dove in to help with editing, references, and anything else that needed to be done. The book would not have occurred without you, Pam. You have made a significant contribution to astronautics that is greatly appreciated.

If you have ideas or suggestions on how we can make the book or the associated website better or more useful, please let us know. In many ways, this book is a new experiment and your feedback is both welcome and appreciated.

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Dedication

Microcosm Press and Microcosm Astronautics Books would like to expressly thank the editors for their vision, the authors for their views, the reviewers and researchers for their help and all for their hard work.

We would like to thank NASA, the US Air Force, US Army, US Navy, European Space Agency, Canadian Space Agency and all the universities, industry and commercial companies that support space programs, for their inspiration.

We would also like to thank the professors that will use this text to teach the next generation for giving us purpose. We would like to thank the small companies that keep on living the dream and the large companies that keep the dream alive, all of whom have been great supporters.

In the end, this book is for us all. We are a small community, in the grand scheme of things, and we take pride in the accomplishments of ourselves and each other. The willingness to share these accomplishments and the knowledge we learn from them is one of our greatest attributes.

So this book is dedicated to us all. May we all live long and prosper!!

Pam Esquinca
Chris Maruyama

1 Introduction

James R. Wertz, ***Microcosm, Inc./USC***

1.1 What is Space Mission Engineering?

1.2 The History of Spaceflight—*Mike Gruntman*
Humble Beginnings; Great Pioneers; Building the Foundation; The Breakthrough to Space; Spacefaring Civilization

1.3 Spaceflight Technology
Getting to Space is Really Hard; Rockets Are a Lousy Way to Get Anywhere; Space Parts Are All Unique; Technology Readiness Levels

1.4 Spaceflight Economics—*John M. Jurist, David M. Livingston*
Space Budgets; Diversity of Opinions; Overview of Economics of Aerospace Systems; General Accepted Accounting Practices (GAAP) and Government Accounting; Why is Space Expensive; How Might Costs Be Minimized?; Other Launch Cost Issues

1.5 The Wide Range of Space Mission Applications

1.6 Sources of More Information

The goal of this book is to lay out the entire process of analyzing and designing space missions—from an initial vague concept or mission objective to designing, building, launching, operating, and eventually deorbiting one or more spacecraft. In the words of former NASA Administrator Dan Goldin, our objective is to do this "faster, better, cheaper" than it has been done before. Although many throughout the space community will argue that this is effectively impossible ("faster, better, cheaper—pick any two" is a common rejoinder), experience has shown that it is possible and that progress is real. Perhaps most important, we **must** find ways to do the job faster, better, and cheaper if we are to accomplish the missions that we want to accomplish within the budgets that are available and enable some of the remarkable benefits that space can bring about. To do this, we need to understand both how and why things are done as they are today, so that we can make the process better tomorrow.

In a sense, this book is intended as a microcosm of the faster, better, cheaper mission engineering process. Our goal with *Space Mission Engineering* is to greatly improve and shorten the preliminary mission design process by making it easier and faster to apply equations and examples and find reference data in time to be of use in a current project. For example, most of the analytic models, numerical tables, and plots throughout the book are immediately available to you on the book website and can be changed to show your specific problem of interest. Some extended discussions of relevant material that is a bit too long for the printed text is also there. Most of the references cited throughout the book are immediately available on the web and nearly all of the remainder are available within a day or so, although, of course, copyright material must be purchased from the owner or an appropriate reseller. We also have a bibliography for each chapter on the website, both so that you can have immediate access to that information, and also so that it can be updated from time-to-time as new books and professional papers appear. Details of what's on the website, how to use it, and other sources of mission engineering information, are described in Sec. 1.6. Note that the website is an adjunct to the book, and not a substitute.

Looking at the organization of the book, Chaps. 1 and 2 provide the philosophical basis and broad context for determining what space mission engineering is all about. Chaps. 3–6 then provide the basic engineering details of defining a space mission, undergoing system trades to find the best available solution (or at least an acceptable one), quantifying the cost and utility of the mission, and undertaking the formal requirements definition process. Chaps. 7–10 look specifically at the space environment, orbit and mission geometry, and orbit and constellation design. Chapters 11–13 then look at financial and legal issues and the fundamental problem of reducing cost and schedule. Part II (Chaps. 14–25) goes through the process of spacecraft and payload design. Part III (Chaps. 26–31) covers launch, operations, and end-of-life. Finally, the appendices and end-pages summarize the numerical data and basic formulas that you will need for preliminary mission design.

1.1 What is Space Mission Engineering?

***Space Mission Engineering* is the definition of mission parameters and refinement of requirements so as to meet the broad and often poorly defined objectives of a space mission in a timely manner at minimum cost and risk. The goal of this book is to give you the processes and tools to do that.**

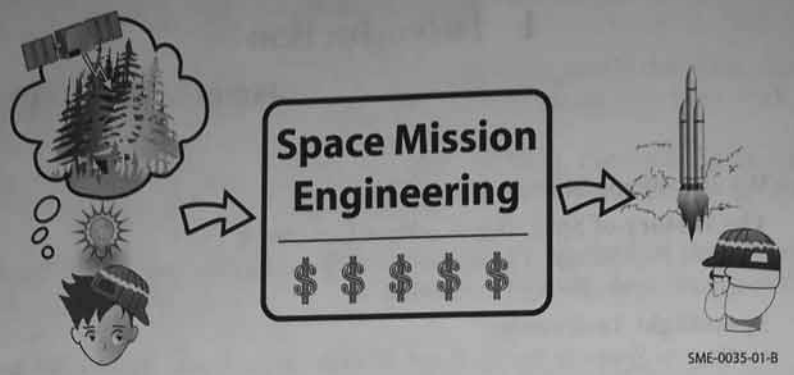

Fig. 1-1. Space Mission Engineering is the Process of Going From Broad Mission Objectives or a Vague Concept to an Operational Mission. Unfortunately many modern space programs perform well, but cost too much and take too long, which is a fundamental problem that is addressed throughout this book.

A good example of the space mission engineering process is the Apollo program that resulted in landing the first humans on the Moon and is one of the major technological achievements of modern times. But how well did it work from an engineering perspective? In its fundamental goal of putting people on the Moon and returning them safely to Earth it was completely successful. In a task that was very high risk, Apollo landed people on the Moon, gave them the time and tools to explore their immediate environment, and returned all of them safely to Earth.* However, our space mission engineering goal is to do this at minimum cost and risk—and we don't really know whether we achieved that. Realistically, we'll settle for reasonable cost and risk.

Like the Apollo program, most modern space missions perform well, and therefore, the space mission engineering job is well done in many ways. However, in another respect, the job is done very badly in that most large space programs today cost too much and take too long (Fig. 1-1). To be more specific, the United States launches approximately 20 space missions per year (including both manned and unmanned, Earth orbiting and interplanetary). The combined US space budget for NASA, DoD, the Intelligence Community, and a few other agencies is on the order of $60 billion/year. (See Sec. 1.5.) This implies that in recent years,

> **Space systems launched by the United States have cost an average of $3 billion/launch. The average time from concept to launch is not well known, but is certainly approaching a decade if not more.**

In my view, that is too much and too long, and fixing it is going to require major changes in how space business is done. Certainly some people within the space community will argue that the cost and time are necessary and appropriate and we will return to this question in Chap. 13. However, irrespective of whether you believe we are doing the job well or badly or, perhaps some combination of both, the goal of this book is to allow you to do it **better**—to create new missions, defined by you, the reader, that are safer, better, more reliable, with greater performance, done more quickly and at lower cost than has been the case in the last decade. In other words, we want space missions to evolve, just as other products in the marketplace evolve.

A part of this evolution includes updating the space mission engineering process from when it was presented 20 years ago in the first edition of *Space Mission Analysis and Design*, known as SMAD [Wertz and Larson, 1991]. The process presented here differs from that in SMAD by adding a timeliness requirement and identifying the people and groups involved, often called the *stakeholders*. There are specific reasons for these changes as discussed in Sec. 3.2.

There is another respect in which the Apollo program was a good example of space mission engineering. Initially, the objectives were broad and had very little definition. President Kennedy, in his proclamation in support of Apollo, did not specify how many people would go on each mission, or how long they would stay, or how much Lunar material they would bring back. The Apollo program itself defined the more detailed objectives, largely by determining what was reasonably feasible within the constraints of the technology. The successive Apollo missions continued to evolve and each landing became more complex and more successful in terms of the areas explored and the material returned. As presented in some detail by Johnson [2002], Apollo was one of the first major examples of systems management and systems engineering at a massive level. Apollo took a vague, poorly defined objective (as it should be) and, using 100's of thousands of people, built the largest, most reliable, and lowest cost per kg rocket ever built, and used that rocket to have 2 people step onto the surface of the Moon only 8.5 years after Kennedy expressed his initial vision. As Johnson calls it, the "Secret of

*Astronauts Gus Grissom, Ed White, and Roger Chaffee were killed in a fire inside the Apollo 1 Command Module during a ground test prior to any of the operational or test flights.

Apollo" was the organizational and engineering process that was able to achieve that.*

In another respect, Apollo wasn't a good example of space mission engineering. It may or may not have been minimum cost and risk, but it certainly wasn't cheap. (Apollo cost well over $150 billion in today's dollars.) What we would like to do today is to do missions much more substantive than Apollo, and for a lot less money. As we will see in Sec. 1.5 and Chap. 2, there is an enormous potential in space and we have only begun to scratch the surface of what can be accomplished for science, for commerce, for the military, for protecting the environment here on Earth, and for starting to use the resources of space for the benefit of mankind. We would like to change that and have missions become faster, cheaper, more useful, and in far greater numbers. All of this requires thinking seriously about both performance **and** cost and finding new ways to do business in space.

What happened? How did we go from having a competent, robust, very active space program to one that is still very competent but is also, for the number of missions launched, dramatically expensive and very slow to respond? At least a portion of the answer probably lies in the *Space Spiral* shown in Fig. 1-2. The ever-increasing cost of space missions leads to longer schedules and fewer missions. This leads to a demand for higher reliability, which, in turn, leads to higher cost, longer schedules, and fewer missions.

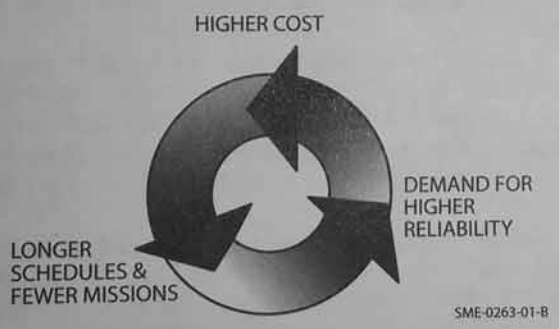

Fig. 1-2. The Space Spiral is a Significant Contributor to Increasing Costs and Longer Schedules.

Chapter 13 provides a detailed assessment of how to solve this problem. A part of the solution is to find ways to reverse the Space Spiral by having a mix of smaller, lower-cost missions and larger, more expensive ones, such that there is less demand for absolute guarantees of success, which are unachievable in the real world. An analogy is a military commander whose troops are on one side of a large canyon and he expects them to be attacked the next day by an enemy force that will move across a bridge over the canyon. The commander sends 3 airplanes to knock out the bridge, when the job could be done by a single plane. It is not that the commander believes that his planes are poorly built or that his pilots are incompetent, it's just that in the real world things go wrong and if you really want that bridge to not be there in the morning, you provide multiple ways of getting the job done. Similarly, in spaceflight, we will achieve a higher level of mission reliability by having multiple, diverse assets available instead of trying to make a single, dramatically expensive asset work perfectly.†

The broad timeline over which space missions are created is described in Sec. 3.4.2. The first phase is Concept Exploration, which is the initial study phase that results in a broad definition of the space mission and an initial estimate of cost, schedule, and performance. While much of this is a vague and "fuzzy" process, it's a critical part of space mission engineering because most of the cost, risk, schedule, and utility of the mission are determined by the end of concept exploration. Therefore, it's important to do this job carefully, thinking about what can be done to reduce the cost and complexity of the mission or increase the utility to the end users and those who will ultimately be asked to pay for the mission. By far the majority of mission concepts die during or at the end of concept exploration. This may be because there is insufficient performance or utility to the end user or because the technology doesn't yet exist to proceed, but it is most often simply because space missions are very expensive and no one is willing to pay to proceed to subsequent, ever more expensive, steps.

Even large, well-defined, active space programs can run into budget overruns and extended schedule delays. The reasons for this, and possible solutions, are described in Secs. 1.3 and 1.4 and in more detail in 13.6. As a result, missions on which hundreds of millions or billions of dollars have been spent are, nonetheless, cancelled from time to time or, when not cancelled, cause major delays in other important programs due to "soaking up" the available resources. This again emphasizes the need for good mission engineering throughout the mission lifecycle. We need to continuously look for ways to constrain cost and schedule growth and to sell the program and keep it sold, if it is still worth doing.

While there are problems with today's space missions, there is also dramatic potential, as described in Sec. 1.5 and Chap. 2. While some missions, such as navigation, communications, and direct broadcast are well-established and routine activities, other missions, such as utilization of space resources, space manufacturing, and

* Many people would argue that the US has lost some key element of the skill or will that was needed to allow Americans to walk on the Moon over 40 years ago and, consequently, the next people to walk on the Moon, and explore it in more depth, will likely be Chinese.

†Both aircraft and ground vehicles are much more reliable than space systems. Launch vehicles are about 90% reliable. We would never use airplanes or automobiles that had a 10% chance of a catastrophic failure that would destroy both the vehicle and its cargo when first used. We can either spend ever more money searching for perfection or find ways to accomplish our space mission objectives in the presence of occasional system failures.

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